Newman-Penrose Formalism Calculations

Stephani et al., Equation (22.29)

Metric

$$ds^2 = -e^{(-b(\rho))}h\left(\rho\right)^2 \mathrm{d}t \otimes \mathrm{d}t + \frac{e^{(-b(\rho))}}{(a\rho - \rho^2 + c)h\left(\rho\right)^2}\mathrm{d}\rho \otimes \mathrm{d}\rho + \frac{1}{h\left(\rho\right)^2}\mathrm{d}y \otimes \mathrm{d}y + \frac{\rho}{h\left(\rho\right)^2}\mathrm{d}y \otimes \mathrm{d}\phi + \frac{\rho}{h\left(\rho\right)^2}\mathrm{d}\phi \otimes \mathrm{d}y + \left(\frac{a\rho + c}{h\left(\rho\right)^2}\right)\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, \rho, y, \phi)$$

Metric Functions

$$E(\rho) = a\rho - \rho^2 + c$$

$$w^2(\rho) = e^{(-b(\rho))}h\left(\rho\right)^2$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{h\left(\rho\right)^2}e^{\left(-\frac{1}{2}b(\rho)\right)}\right)dt + \left(-\frac{\sqrt{2}e^{\left(-\frac{1}{2}b(\rho)\right)}}{2\sqrt{a\rho - \rho^2 + c}h\left(\rho\right)}\right)d\rho$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{h\left(\rho\right)^2}e^{\left(-\frac{1}{2}b(\rho)\right)}\right)dt + \left(\frac{\sqrt{2}e^{\left(-\frac{1}{2}b(\rho)\right)}}{2\sqrt{a\rho - \rho^2 + c}h\left(\rho\right)}\right)d\rho$$

$$m = \left(\frac{\sqrt{2}}{2h\left(\rho\right)}\right)dy + \left(\frac{\sqrt{2}\rho}{2h\left(\rho\right)} - \frac{i\sqrt{2}\sqrt{a\rho - \rho^2 + c}}{2h\left(\rho\right)}\right)d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,h\left(\rho\right)} \right)\, dy + \left(\frac{\sqrt{2}\rho}{2\,h\left(\rho\right)} + \frac{i\,\sqrt{2}\sqrt{a\rho - \rho^2 + c}}{2\,h\left(\rho\right)} \right)\, d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}e^{\left(\frac{1}{2}\,b(\rho)\right)}}{2\left|h\left(\rho\right)\right|} \right)\, dt + \left(-\frac{1}{2}\,\sqrt{2}\sqrt{a\rho - \rho^2 + c}e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right) \right)\, d\rho$$

$$n = \left(-\frac{\sqrt{2}e^{\left(\frac{1}{2}\,b(\rho)\right)}}{2\left|h\left(\rho\right)\right|} \right)\, dt + \left(\frac{1}{2}\,\sqrt{2}\sqrt{a\rho - \rho^2 + c}e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right) \right)\, d\rho$$

$$m = \left(\frac{\sqrt{2}a\rho h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} - \frac{\sqrt{2}\rho^2 h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} + \frac{i\,\sqrt{2}\rho h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}} + \frac{\sqrt{2}c h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} \right)\, dy + \left(-\frac{i\,\sqrt{2}h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}} \right)\, d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}a\rho h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} - \frac{\sqrt{2}\rho^2 h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} - \frac{i\,\sqrt{2}\rho h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}} + \frac{\sqrt{2}c h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} \right)\, dy + \left(\frac{i\,\sqrt{2}h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}} \right)\, d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}e^{\left(\frac{1}{2}\,b(\rho)\right)}}{2\left|h\left(\rho\right)\right|}\,\partial_t - \frac{1}{2}\,\sqrt{2}\sqrt{a\rho - \rho^2 + c}e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right)\,\partial_\rho$$

$$\Delta = -\frac{\sqrt{2}e^{\left(\frac{1}{2}\,b(\rho)\right)}}{2\left|h\left(\rho\right)\right|}\,\partial_t + \frac{1}{2}\,\sqrt{2}\sqrt{a\rho - \rho^2 + c}e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right)\,\partial_\rho$$

$$\delta = \frac{\sqrt{2}a\rho h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} - \frac{\sqrt{2}\rho^2 h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} + \frac{i\,\sqrt{2}\rho h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}} + \frac{\sqrt{2}c h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)}\,\partial_y - \frac{i\,\sqrt{2}h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}a\rho h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} - \frac{\sqrt{2}\rho^2 h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)} - \frac{i\,\sqrt{2}\rho h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}} + \frac{\sqrt{2}c h\left(\rho\right)}{2\left(a\rho - \rho^2 + c\right)}\,\partial_y + \frac{i\,\sqrt{2}h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^2 + c}}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2} a\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2}\rho^{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2} a e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2}\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2} c e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^{2} + c}}$$

$$\sigma = -\frac{i\,\sqrt{2} a\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\left(a\rho - \rho^{2} + c\right)} + \frac{i\,\sqrt{2}\rho^{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\left(a\rho - \rho^{2} + c\right)} - \frac{\sqrt{2} a e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2}\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} - \frac{i\,\sqrt{2} c e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\left(a\rho - \rho^{2} + c\right)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2} a\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2}\rho^{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2} a e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2}\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2} c e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{2\,\sqrt{a\rho - \rho^{2} + c}}$$

$$\lambda = \frac{i\,\sqrt{2} a\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\left(a\rho - \rho^{2} + c\right)} - \frac{i\,\sqrt{2}\rho^{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\left(a\rho - \rho^{2} + c\right)} - \frac{\sqrt{2} a e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2}\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} + \frac{i\,\sqrt{2} c e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)}{4\left(a\rho - \rho^{2} + c\right)}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2} a\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right) \frac{\partial}{\partial\rho} b\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2}\rho^{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right) \frac{\partial}{\partial\rho} b\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2} c e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right) \frac{\partial}{\partial\rho} b\left(\rho\right)}{8\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2} a\rho e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} + \frac{\sqrt{2}\rho^{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} - \frac{\sqrt{2} c e^{\left(\frac{1}{2}\, b(\rho)\right)} \frac{\partial}{\partial\rho} h\left(\rho\right)}{4\,\sqrt{a\rho - \rho^{2} + c}} - \frac{1}{8} i\,\sqrt{2} e^{\left(\frac{1}{2}\, b(\rho)\right)} h\left(\rho\right)$$

$$\gamma = \frac{\sqrt{2}a\rho e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)}{8\,\sqrt{a\rho-\rho^2+c}} - \frac{\sqrt{2}\rho^2 e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)}{8\,\sqrt{a\rho-\rho^2+c}} + \frac{\sqrt{2}ce^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)}{8\,\sqrt{a\rho-\rho^2+c}} - \frac{\sqrt{2}a\rho e^{\left(\frac{1}{2}\,b(\rho)\right)}\frac{\partial}{\partial\rho}h\left(\rho\right)}{4\,\sqrt{a\rho-\rho^2+c}} + \frac{\sqrt{2}\rho^2 e^{\left(\frac{1}{2}\,b(\rho)\right)}\frac{\partial}{\partial\rho}h\left(\rho\right)}{4\,\sqrt{a\rho-\rho^2+c}} - \frac{\sqrt{2}ce^{\left(\frac{1}{2}\,b(\rho)\right)}\frac{\partial}{\partial\rho}h\left(\rho\right)}{4\,\sqrt{a\rho-\rho^2+c}} + \frac{1}{8}i\,\sqrt{2}e^{\left(\frac{1}{2}\,b(\rho)\right)}h\left(\rho\right)$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = {} & \frac{1}{2}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{2}\,\rho^2 e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{8}\,ae^{b(\rho)}h\left(\rho\right)^2\frac{\partial}{\partial\rho}b\left(\rho\right) + \frac{1}{4}\,\rho e^{b(\rho)}h\left(\rho\right)^2\frac{\partial}{\partial\rho}b\left(\rho\right) \\ & + \frac{1}{2}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - a\rho e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^2 + \rho^2 e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^2 + \frac{1}{2}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial^2}{(\partial\rho)^2}h\left(\rho\right) - \frac{1}{2}\,\rho^2 e^{b(\rho)}h\left(\rho\right)\frac{\partial^2}{(\partial\rho)^2}h\left(\rho\right) \\ & + \frac{1}{2}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - ce^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^2 + \frac{1}{2}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial^2}{(\partial\rho)^2}h\left(\rho\right) + \frac{1}{8}\,e^{b(\rho)}h\left(\rho\right)^2\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = {} & -\frac{1}{8}\,a\rho e^{b(\rho)}h\left(\rho\right)^2\frac{\partial^2}{(\partial\rho)^2}b\left(\rho\right) + \frac{1}{8}\,\rho^2 e^{b(\rho)}h\left(\rho\right)^2\frac{\partial^2}{(\partial\rho)^2}b\left(\rho\right) - \frac{1}{4}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{4}\,\rho^2 e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) \\ & - \frac{1}{16}\,ae^{b(\rho)}h\left(\rho\right)^2\frac{\partial}{\partial\rho}b\left(\rho\right) + \frac{1}{8}\,\rho e^{b(\rho)}h\left(\rho\right)^2\frac{\partial}{\partial\rho}b\left(\rho\right) - \frac{1}{8}\,ce^{b(\rho)}h\left(\rho\right)^2\frac{\partial^2}{(\partial\rho)^2}b\left(\rho\right) - \frac{1}{4}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{4}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial^2}{(\partial\rho)^2}h\left(\rho\right) \\ & - \frac{1}{4}\,\rho^2 e^{b(\rho)}h\left(\rho\right)\frac{\partial^2}{(\partial\rho)^2}h\left(\rho\right) + \frac{1}{4}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{2}\,\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{4}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial^2}{(\partial\rho)^2}h\left(\rho\right) + \frac{1}{16}\,e^{b(\rho)}h\left(\rho\right)^2\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{2}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{2}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{8}\,ae^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) + \frac{1}{4}\,\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right)$$
$$+ \frac{1}{2}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - a\rho e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \rho^{2}e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \frac{1}{2}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right) - \frac{1}{2}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right)$$
$$+ \frac{1}{2}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - ce^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \frac{1}{2}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right) + \frac{1}{8}\,e^{b(\rho)}h\left(\rho\right)^{2}$$

$$\Lambda = \frac{1}{24}\,a\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial^{2}}{(\partial\rho)^{2}}b\left(\rho\right) - \frac{1}{24}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial^{2}}{(\partial\rho)^{2}}b\left(\rho\right) + \frac{1}{12}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{12}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{48}\,ae^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right)$$
$$- \frac{1}{24}\,\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) + \frac{1}{24}\,ce^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial^{2}}{(\partial\rho)^{2}}b\left(\rho\right) + \frac{1}{12}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{6}\,a\rho e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \frac{1}{6}\,\rho^{2}e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \frac{1}{12}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right)$$
$$- \frac{1}{12}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right) + \frac{1}{12}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{6}\,\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{6}\,ce^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \frac{1}{12}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right) + \frac{1}{16}\,e^{b(\rho)}h\left(\rho\right)^{2}$$

Weyl Tensor Components

$$\Psi_{0} = \frac{1}{8}\,ae^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) - \frac{1}{4}\,\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) + \frac{1}{4}i\,\sqrt{a\rho-\rho^{2}+c}e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) - \frac{1}{4}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{2}\,\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{2}i\,\sqrt{a\rho-\rho^{2}+c}e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right)$$

$$\Psi_{1} = 0$$

$$\Psi_{2} = -\frac{1}{12}\,a\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial^{2}}{(\partial\rho)^{2}}b\left(\rho\right) + \frac{1}{12}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial^{2}}{(\partial\rho)^{2}}b\left(\rho\right) - \frac{1}{6}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{6}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{24}\,ae^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right)$$
$$+ \frac{1}{12}\,\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) - \frac{1}{12}\,ce^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial^{2}}{(\partial\rho)^{2}}b\left(\rho\right) - \frac{1}{6}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}b\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{3}\,a\rho e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} - \frac{1}{3}\,\rho^{2}e^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2}$$
$$+ \frac{1}{3}\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right) - \frac{1}{3}\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right) + \frac{1}{12}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) - \frac{1}{6}\,\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{3}\,ce^{b(\rho)}\frac{\partial}{\partial\rho}h\left(\rho\right)^{2} + \frac{1}{3}\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial^{2}}{(\partial\rho)^{2}}h\left(\rho\right)$$

$$\Psi_{3} = 0$$

$$\Psi_{4} = -\frac{i\,a\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right)}{4\,\sqrt{a\rho-\rho^{2}+c}} + \frac{i\,\rho^{2}e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right)}{4\,\sqrt{a\rho-\rho^{2}+c}} + \frac{1}{8}\,ae^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) - \frac{1}{4}\,\rho e^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right) - \frac{i\,ce^{b(\rho)}h\left(\rho\right)^{2}\frac{\partial}{\partial\rho}b\left(\rho\right)}{4\,\sqrt{a\rho-\rho^{2}+c}}$$
$$+ \frac{i\,a\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right)}{2\,\sqrt{a\rho-\rho^{2}+c}} - \frac{i\,\rho^{2}e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right)}{2\,\sqrt{a\rho-\rho^{2}+c}} - \frac{1}{4}\,ae^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{1}{2}\,\rho e^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right) + \frac{i\,ce^{b(\rho)}h\left(\rho\right)\frac{\partial}{\partial\rho}h\left(\rho\right)}{2\,\sqrt{a\rho-\rho^{2}+c}}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"